Unit ILO	Contributing assessment parts	Fail						3rd Class			2:2 (Lower Second) Class			2:1 (Lower Second) Class			First Class			Exceptional First Class		
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
		0	7	15	22	29	35	42	45	48	52	55	58	62	65	68	72	75	78	83	94	100
ILO 1: **Describe and select the fundamental elements of an electric drive system and the benefits that these elements bring to a particular application.**	Parts 1 and 2	**No/unsufficient evidence of understanding of drive technologies or how these interface with one another**						Completes at least some of the requirements with some errors replicating some of the design studies taught in the lab exercises for at least 1 of the 2 elements of the drive (machine/converter)			Completes many of the requirements correctlty of the design problem for at least 1 elements of the drive (system/ converter)			Completes all of the requirements correctly for at least 1 element of the design problem (machine/ component)			Completes all of the requirements correctly for at least 2 element of the design problem (machine/ component)			**Part 3 of the assessment provides evidence to access these criteria below** Completes all of the requirements correctly for all elements of the design problem and extends these analyses proposing new context-appropriate solutions/ideas		
ILO 2: **Source, extract and use appropriate data to populate relevant models for design evaluation.**	Parts 1 and 2	**Significant or hazardous evidence of incorrect use of datasheets/ design data for problem analysis/ simulation**						Identifies data for relevant components to populate simulation and/or uses relevant data from design problem in simulation/ analysis for at least 1 element of the drive			Extracts simple data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation correctly in most cases for at least 1 element of the drive.			Extracts data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation appropriately in all cases for at least 1 element of the drive .			Extracts data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation appropriately in all cases for 2 elements of the drive .			**Part 3 of the assessment provides evidence to access these criteria below** Extracts new data from externally sourced data sheets/design problem data and exploits their benefits in design problem using developed tools to quantify and demosntrate confidence in their ideas		
ILO 3: **Design and evaluate an electric drive based upon an appropriate converter and machine topology for the various elements of the system using a mixture of analytical techniques and simulation tools across multiple domains for example: electrical, mechanical and thermal.**	Parts 1, 2 and 3	**Little evidence or incorrect application of appropriate analysis/simulation method resulting in no or very poorly formed outcomes.**						Undertakes an analytical method and simulation for the design problem. Selects an appropriate simulation method for the task being undertaken for each task. Builds and uses simulations to test outcomes appropriately and provide cross validation to build confidence in design tools.			**Part 3 of the assessment provides evidence to access these criteria below** Uses developed design tools (analytical framewortk and simulation) to explore additional operating points and come to simple conclusions.			As 2:2 with clear evidence and reporting of more context aware exploration in at least one element of the design problem (system, converter) as it is presented. Findings all appropriate.			Demonstrates a clear line of thought in producing a solution to the design problem. As for 2:1 but with a clear demonstration of exploring the design problem with creative selection/estimation of design deviations for further exploration or with a clear design intent and reflection.			Providing an analysis that undertakes a thorough context-aware design analysis over multiple/all elements of the problem and extends beyond the original design specification and in the context of the application/ system problem defintion and across multiple domains.		
ILO 4: **Reflect on the strengths and weaknesses of a design solution with respect to the application requirements using appropriate evaluaton metrics.**	Parts 1, 2 and 3	**No/Poor documentation of and reflection on design process and outcomes.**						Uses results to detemine engineering requirements that satisfies a the simple required design tasks with little reflection			**Part 3 of the assessment provides evidence to access these criteria below** Produces results that satisfy many of the required design tasks as presented but at additional operating points with limited reflection.			Produces results that satisfy most of the required design tasks with discussion of trade-offs required and/or simplifications/ assumptions made, or details added and reflection on the process.			Clear reporting of the design path and outcomes and indentification of path for further exploration were there more time and/or clear reflection on the areas of uncertainty in the results.			Clear evidence and concise reporting of well thought out multi-domain design exploration and refelction on the outcomes in the context of the system /application rather than just the technology.		